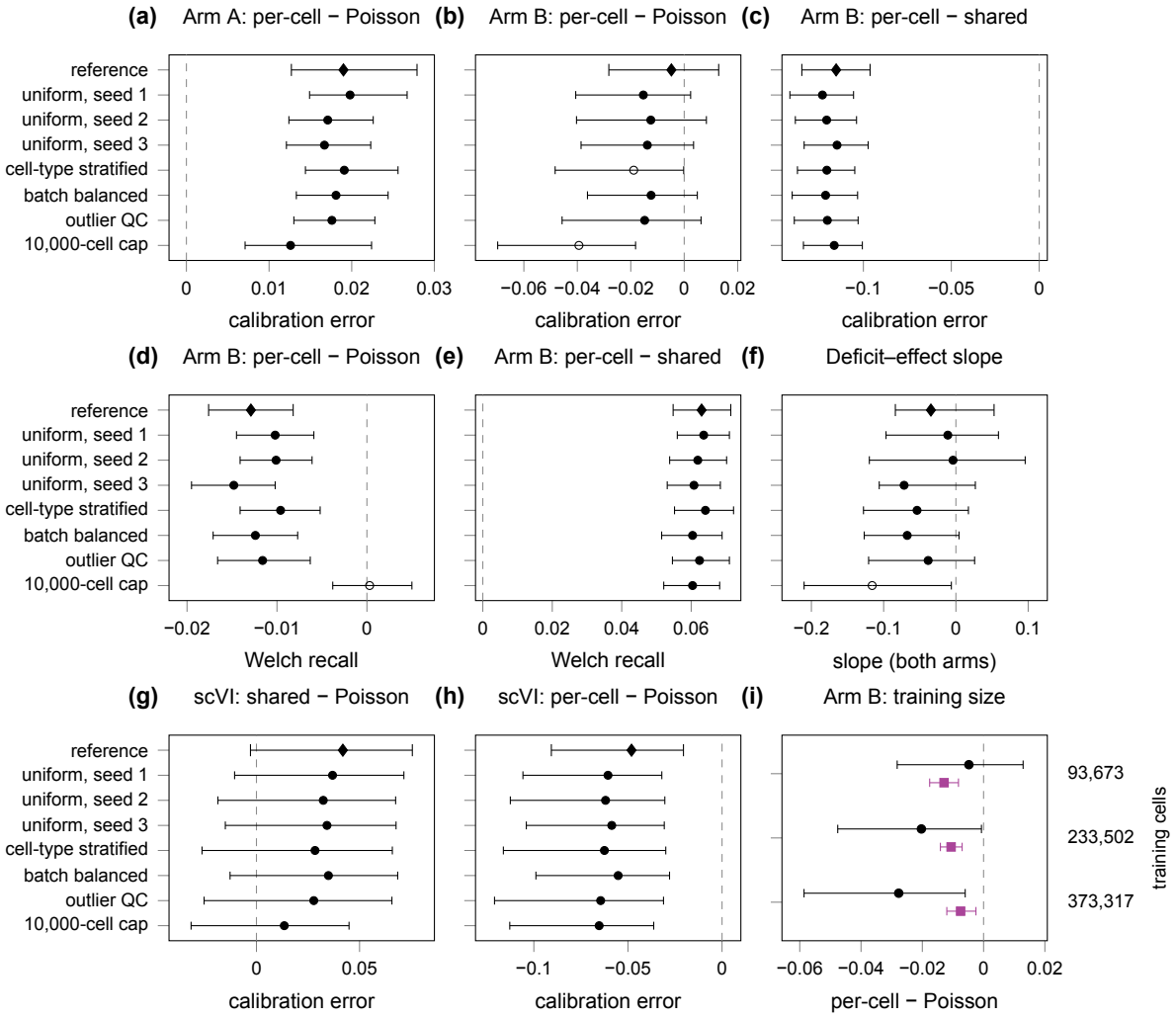
(a) Arm A: per-cell − Poisson
(b) Arm B: per-cell − Poisson
(c) Arm B: per-cell − shared
reference
uniform, seed 1
uniform, seed 2
uniform, seed 3
cell-type stratified
batch balanced
outlier QC
10,000-cell cap
0
0.01
0.02
0.03
−0.06
−0.04
−0.02
0
0.02
−0.1
−0.05
0
calibration error
(d) Arm B: per-cell − Poisson
(e) Arm B: per-cell − shared
(f) Deficit–effect slope
−0.02
−0.01
0
0
0.02
0.04
0.06
−0.2
−0.1
0
0.1
Welch recall
slope (both arms)
(g) scVI: shared − Poisson
(h) scVI: per-cell − Poisson
(i) Arm B: training size
0.05
−0.1
−0.05
0
−0.06
−0.04
−0.02
0.02
per-cell − Poisson
93,673
233,502
373,317
training cells